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1. Introduction

> As the new typeface for our diverse and global brand, Plex® is just as important as our name or our logo. It fine-tunes the tone of our words. It represents who we are and what we believe—as a company and as designers.
> (See https://www.ibm.com/plex/plexness/.)

IBM Plex is a bit of a strange font family to me. I think it looks nice—I especially love how reminiscent it is of the IBM Selectric and Composer machines—but, to be honest, it's *extremely* corporate to me. This is because, well, it just *is* corporate; IBM Plex is a *proudly* corporate typeface because it was made for a giant tech company's brand identity.

However, there are times when I might want to use this.

2. Standard demo

The following text is taken from Lieb, E., Schultz, T., Mattis, D. (2004).[1] Please, go read the actual paper, it's great.

2.1. Formulation

The first model consists of N spin 1/2's (N even) arranged in a row and having only the nearest neighbor interactions. It is

$$H_\gamma = \sum_i \left[(1+\gamma)S_i^x S_{i+1}^x + (1-\gamma)S_i^y S_{i+1}^y\right],$$

where γ is a parameter characterizing the usual degree of anisotropy in the xy-plane. Because the Hamiltonian only involves the $x-$ and $y-$ components of the spin operators, we call this model the XY model.

[1] Two Soluble Models of an Antiferromagnetic Chain. In: Nachtergaele, B., Solovej, J.P., Yngvason, J. (eds) Condensed Matter Physics and Exactly Soluble Models. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06390-3_35

To solve the XY model, we first introduce the raising and lowering operators

$$a_i^\dagger = S_i^x + \mathrm{i}\, S_i^y \text{ and } a_i = S_i^x - \mathrm{i} S_i^y$$

in terms of which the Pauli spin operators are

$$S_i^x = \left(a_i^\dagger + a_i\right)/2;\; S_i^y = \left(a_i^\dagger - a_i\right)/2\mathrm{i};\; S_i^z = a_i^\dagger\, a_i - \frac{1}{2}$$

and the Hamiltonian is

$$H_\gamma = \frac{1}{2} \sum_i \left[\left(a_i^\dagger\, a_{i+1} + \gamma\, a_i^\dagger\, a_{i+1}^\dagger\right) + \text{h.c.}\right].$$

The ground state Ψ_0 is the state with no elementary excitations:

$$\eta_k\, \Psi_0 = 0\,,\ \text{all } k\,.$$

The ground-state energy, according to (A-12) is

$$E_0 = -\frac{1}{2} \sum_k \Lambda_k\,.$$

In the limit $N \to \infty$, the sum can be replaced by an integral giving

$$\begin{aligned}\frac{E_0}{N} &= -\frac{1}{4\pi} \int_{-\pi}^{+\pi} \left[1 - \left(1 - \gamma^2\right) \sin^2 k\right]^{1/2} dk \\ &= -\frac{1}{\pi}\, \mathcal{E}\left(1 - \gamma^2\right),\end{aligned}$$

where $\mathcal{E}\left(k^2\right)$ is one of the complete elliptic integrals (13).